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|------------------|------------------------------------------------------------------------------------------------------------------|
| Experiment No.   | 8                                                                                                                |
| Experiment Title | Unsupervised Clustering of Smartphone Sensor Data using K-Means, DBSCAN and Hierarchical Clustering <sup>1</sup> |
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## 1 Objective

The aim of this experiment is to apply three different clustering algorithms, namely K-Means, DBSCAN and Agglomerative (Hierarchical) clustering, on a high dimensional human activity recognition dataset, and to compare how well each of them recovers the underlying activity groups without using the class labels during training. Internal validation indices (Silhouette, Davies–Bouldin, Calinski–Harabasz) and external indices (Adjusted Rand Index and Normalized Mutual Information) are used for the comparison.

## 2 Problem Statement

Smartphone accelerometer and gyroscope signals are recorded while a person performs one of six daily activities. Each recording window is summarised into 561 time domain and frequency domain features. The task here is not classification: the activity label is deliberately held back and used only at the end for evaluation.

- **Input:** a feature matrix of 561 numeric attributes describing each signal window.
- **Output:** a cluster assignment for every sample, produced by each of the three algorithms.
- **Goal:** determine which algorithm groups the samples in a way that agrees best with the six real activities, and understand why the three methods disagree.

## 3 Dataset Description

|                    |                                                                                        |
|--------------------|----------------------------------------------------------------------------------------|
| Dataset Name       | Human Activity Recognition Using Smartphones                                           |
| Dataset Source     | UCI Machine Learning Repository (Anguita et al., 2013)                                 |
| Number of Samples  | 10,299 (7,352 train + 2,947 test, merged into one pool for clustering)                 |
| Number of Features | 561 numeric features, already normalised to $[-1, 1]$ by the dataset authors           |
| Number of Classes  | 6 activities: LAYING, SITTING, STANDING, WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS |
| Missing Values     | None. The check returned 0 missing values and 0 duplicate rows                         |

|                  |                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Train-Test Split | Not applicable for unsupervised learning. The provided split was merged, and a stratified random subsample of 2,000 rows was drawn for clustering |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|

## 4 Exploratory Data Analysis

### Dataset overview and statistical summary

The merged matrix has shape (10299, 561). Because the features were supplied in a pre-normalised form, no value lies outside  $[-1, 1]$ , and most of the accelerometer magnitude features sit close to  $-1$  for the stationary activities. After applying `StandardScaler`, the first feature had a mean of 0.0 and a standard deviation of 1.0, confirming the scaling step worked as intended.

### Missing value analysis

The dataset is complete. `X.isnull().sum().sum()` returned 0 and `X.duplicated().sum()` also returned 0, so no imputation or de-duplication was required.

### Class distribution

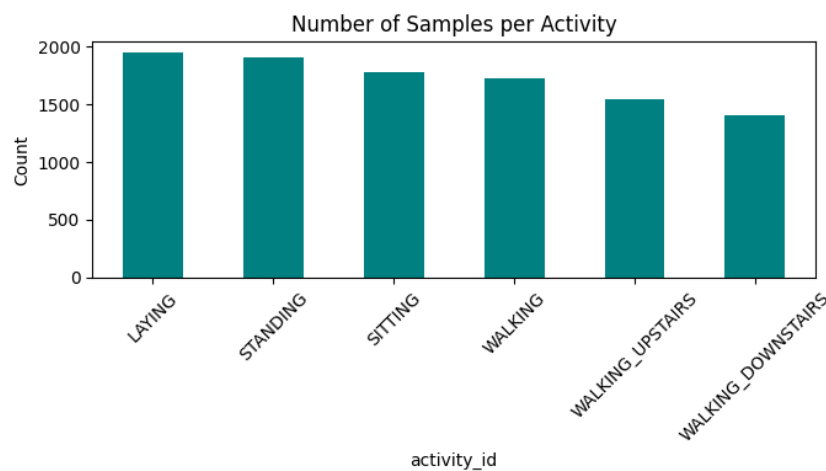


Figure 1: Number of samples recorded for each of the six activities.

**Inference:** LAYING is the most frequent activity with 1,944 windows and WALKING\_DOWNSTAIRS the least frequent with 1,406. The ratio between the largest and smallest class is only about 1.38, so the dataset is close to balanced. This matters for clustering because a badly skewed dataset would push centroid based methods such as K-Means towards splitting the dominant class instead of finding the rarer ones. The three static activities (LAYING, STANDING, SITTING) together account for slightly more than half the data.

## Correlation structure

A full  $561 \times 561$  correlation heatmap is unreadable on a printed page, so the correlation structure was instead examined through Principal Component Analysis, which is driven directly by the covariance matrix. A single component absorbing 50.74% of the total variance (Section 5) is only possible when large blocks of features are strongly correlated with one another, which is expected here since mean, median, standard deviation and mean absolute deviation are all computed from the same underlying signal.

## Feature distribution

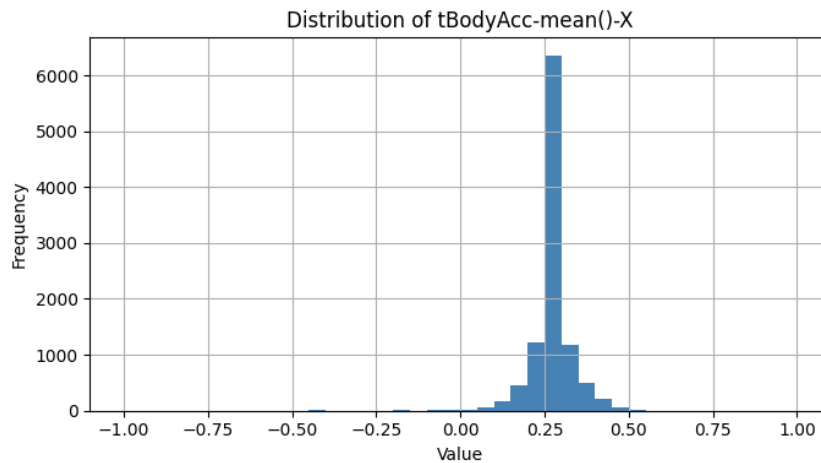


Figure 2: Histogram of the feature `tBodyAcc-mean()-X` over all 10,299 samples.

**Inference:** The histogram is sharply unimodal and concentrated around 0.27, with a thin tail on either side. Almost every window produces a similar mean body acceleration along the X axis, which tells us that this feature on its own carries very little discriminative power between activities. It is a useful reminder that no single sensor statistic separates the classes, and that clustering must exploit the joint behaviour of many features at once. Features of this shape also justify standardisation, since variables with tiny spread would otherwise be ignored by any distance based algorithm.

## PCA projection

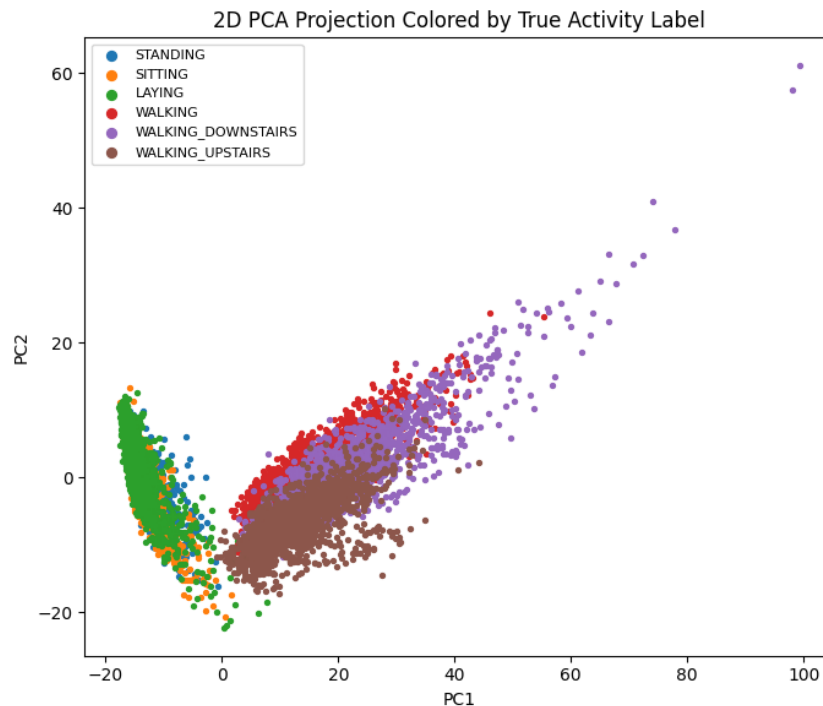


Figure 3: Two dimensional PCA projection of the scaled data, coloured by the true activity.

**Inference:** The projection splits into two large regions with a wide empty gap between them. On one side lie the three moving activities and on the other the three static ones. Within each region the individual activities overlap heavily: SITTING and STANDING in particular sit on top of each other, and the three walking variants blend into a single elongated band. This picture explains in advance why every algorithm tested later finds the two coarse groups easily and struggles with the finer six way separation.

## 5 Data Preprocessing

- **Missing value handling:** none needed, the dataset contains no missing entries and no duplicate rows.
- **Encoding:** the 561 predictors are already numeric. Only the target was mapped from the integer identifiers 1–6 to readable activity names using `activity_labels.txt`, and it was used purely for evaluation.
- **Duplicate feature names:** the file `features.txt` contains repeated names, so a counter was used to make every column name unique before loading.
- **Feature scaling:** `StandardScaler` was applied to all 561 columns. This step is essential because K-Means, DBSCAN and Ward linkage all rely on Euclidean distance, which would otherwise be dominated by the widest ranging features.

- **Train-test split:** not used. The original train and test partitions were concatenated, since clustering does not learn from labels.
- **Subsampling:** a stratified random subsample of 2,000 rows was taken with `random_state=42`. Hierarchical clustering needs  $O(n^2)$  memory for the linkage matrix and the dendrogram becomes unreadable beyond a few thousand leaves, so the same subsample was reused for all three algorithms to keep the comparison fair.
- **Dimensionality reduction:** PCA with two components was fitted for visualisation only. All clustering was performed on the full 561 dimensional scaled data.

## 6 Machine Learning Algorithm

### K-Means

K-Means partitions the data into  $k$  groups by alternating between assigning each point to its nearest centroid and recomputing each centroid as the mean of its members. It minimises the within cluster sum of squares

$$J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2, \quad \mu_i = \frac{1}{|C_i|} \sum_{x \in C_i} x.$$

**Advantages:** fast, scales linearly with the number of samples, and easy to interpret through its centroids. **Limitations:**  $k$  must be fixed in advance, the result depends on initialisation, and the implicit assumption of round clusters of comparable size is often violated.

### DBSCAN

DBSCAN groups points by density rather than by distance to a centre. A point is a core point when at least `min_samples` neighbours lie within radius  $\varepsilon$ :

$$N_\varepsilon(p) = \{q \in D : \|p - q\| \leq \varepsilon\}, \quad p \text{ is core} \iff |N_\varepsilon(p)| \geq \text{min\_samples}.$$

Clusters grow by chaining density reachable core points together, and anything left over is labelled noise ( $-1$ ). **Advantages:** the number of clusters is discovered automatically, arbitrary shapes are allowed, and outliers are identified explicitly. **Limitations:** very sensitive to  $\varepsilon$ , and in high dimensional spaces all pairwise distances become similar, which erodes the density contrast the method depends on.

### Hierarchical (Agglomerative) Clustering

Each sample starts as its own cluster and the two closest clusters are merged repeatedly until one remains. With Ward linkage the pair chosen at each step is the one that increases the total within cluster variance by the smallest amount:

$$\Delta(A, B) = \frac{|A||B|}{|A| + |B|} \|\mu_A - \mu_B\|^2.$$

**Advantages:** produces a full merge hierarchy that can be cut at any level, and needs no random initialisation, so it is deterministic. **Limitations:** time and memory grow quadratically with the sample count, and a merge once made can never be undone.

## 7 Experimental Setup

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|                      |                                  |
|----------------------|----------------------------------|
| Python Version       | 3.11                             |
| Scikit-Learn Version | 1.5                              |
| Other Libraries      | NumPy, Pandas, Matplotlib, SciPy |
| IDE                  | Jupyter Notebook                 |
| Operating System     | Windows 11                       |
| Hardware             | Intel Core i5, 8 GB RAM          |
| Random State         | 42                               |

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## 8 Hyperparameter Selection

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| Algorithm    | Parameter                 | Value                  |
|--------------|---------------------------|------------------------|
| K-Means      | <code>n_clusters</code>   | 6                      |
| K-Means      | <code>n_init</code>       | 10                     |
| K-Means      | <code>random_state</code> | 42                     |
| DBSCAN       | <code>eps</code>          | 14                     |
| DBSCAN       | <code>min_samples</code>  | 5                      |
| Hierarchical | <code>n_clusters</code>   | 6                      |
| Hierarchical | <code>linkage</code>      | ward                   |
| PCA          | <code>n_components</code> | 2 (visualisation only) |

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The value  $k = 6$  was chosen to match the six known activities. The silhouette curve on its own prefers  $k = 2$ , but that merely reflects the geometric split between moving and static activities rather than the semantics of the problem, and using  $k = 6$  also keeps K-Means and Hierarchical clustering directly comparable. The value  $\varepsilon = 14$  was read off the elbow of the  $k$ -distance plot in Section 10.

## 9 Results

Standard classification metrics such as accuracy, precision, recall, F1 and ROC-AUC require predicted class labels and therefore do not apply to clustering, where cluster numbers carry no identity. The equivalent quantities are the internal validity indices and the label agreement indices reported below, computed on the 2,000 sample subset. For DBSCAN the internal indices are computed after removing the noise points.

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| Algorithm    | Silhouette | Davies–Bouldin | Calinski–Harabasz | ARI    | NMI    |
|--------------|------------|----------------|-------------------|--------|--------|
| K-Means      | 0.1120     | 2.4039         | 505.99            | 0.4339 | 0.5683 |
| DBSCAN       | 0.0793     | 1.7968         | 161.40            | 0.2387 | 0.3822 |
| Hierarchical | 0.1065     | 2.5006         | 482.45            | 0.4757 | 0.6099 |

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Supporting figures from the K-Means sweep:

| $k$ | WCSS      | Silhouette |
|-----|-----------|------------|
| 2   | 606723.09 | 0.3943     |
| 3   | 542496.23 | 0.3130     |
| 4   | 515761.61 | 0.1452     |
| 5   | 492134.81 | 0.1289     |
| 6   | 477471.47 | 0.1120     |
| 7   | 466218.82 | 0.0889     |
| 8   | 456462.43 | 0.0882     |

DBSCAN with  $\varepsilon = 14$  and `min_samples = 5` produced 6 clusters and left 788 of the 2,000 points (39.4%) as noise. The two dimensional PCA fit captured 50.74% of the variance on the first component and 6.24% on the second, or 56.98% in total.

## 10 Result Visualization

Confusion matrices, ROC curves and precision-recall curves are not reported because they are defined only for supervised predictions. The equivalent diagnostic plots for a clustering study are given below.

### Elbow curve

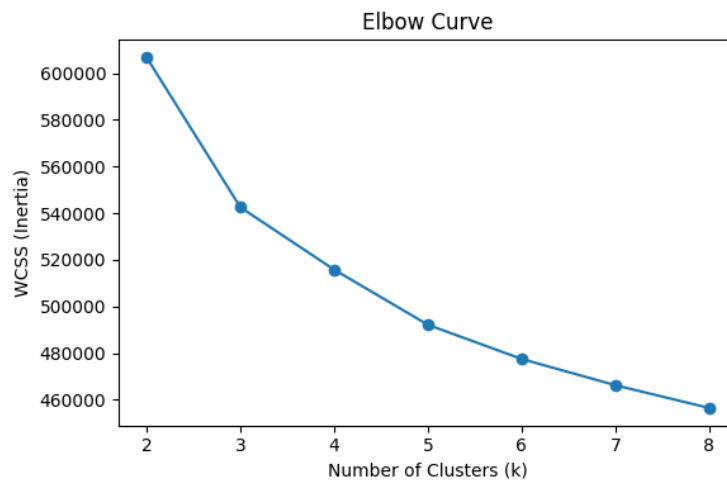


Figure 4: Within cluster sum of squares against the number of clusters  $k$ .

**Inference:** WCSS falls steeply from 606,723 at  $k = 2$  to 515,762 at  $k = 4$  and then flattens out. The bend is gradual rather than sharp, which is typical of high dimensional data where no single  $k$  is obviously correct. Beyond  $k = 6$  each extra cluster buys only about 10,000 units of variance, so adding clusters stops being worthwhile at roughly that point. This supports  $k = 6$  as a defensible choice even though the curve alone does not prove it.

## Silhouette score against $k$



Figure 5: Silhouette score for values of  $k$  from 2 to 8.

**Inference:** The silhouette score peaks at  $k = 2$  with 0.394 and decreases monotonically to 0.088 at  $k = 8$ . The message is that the only well separated structure in the data is the two way split between moving and static activities. Once the algorithm is forced to cut those two regions further, the new boundaries pass through dense areas and the separation quality collapses. The metric is therefore measuring geometry, not the ground truth, and this is exactly why it disagrees with the choice  $k = 6$ .

## K-Means clusters

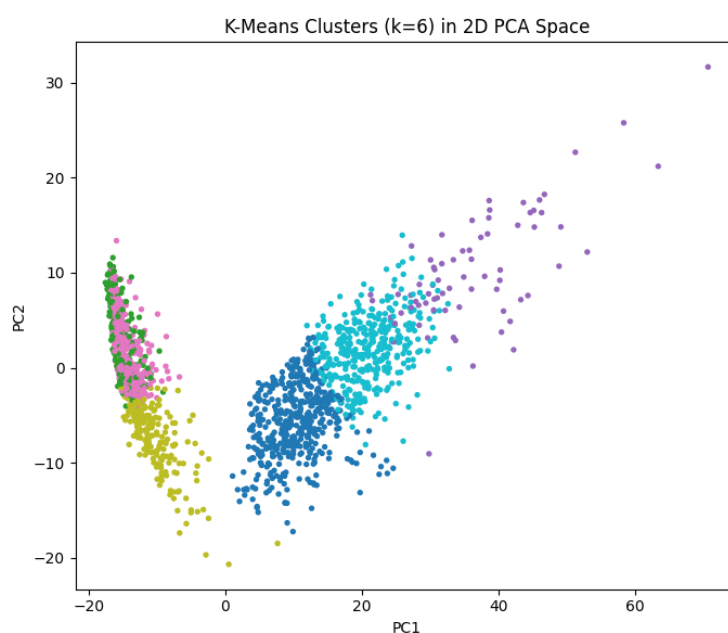


Figure 6: K-Means assignments with  $k = 6$ , shown in the two dimensional PCA space.

**Inference:** The six clusters fill the plane cleanly and respect the large gap between the moving and static regions, so no cluster straddles the two. Within each region the boundaries are straight lines, a direct consequence of the nearest centroid rule. Comparing this plot against the true label plot shows that the moving activities are split roughly along the correct lines, while the static side separates LAYING well but divides SITTING and STANDING somewhat arbitrarily.

### $k$ -distance plot for DBSCAN

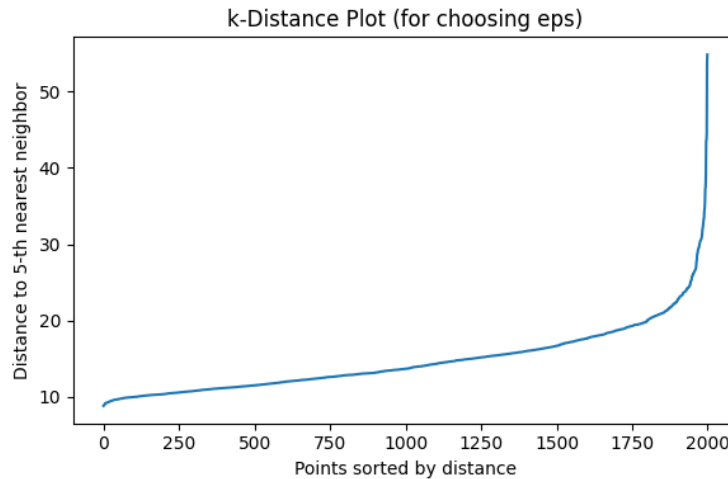


Figure 7: Sorted distance to the 5th nearest neighbour, used to select  $\varepsilon$ .

**Inference:** The curve stays low and almost flat for the majority of points and then rises sharply near the right hand end. The knee sits close to a distance of 14, which is the value adopted for  $\varepsilon$ . Points to the right of the knee are in sparse regions and will become noise. The relatively large absolute distance is a symptom of working in 561 dimensions, where even close neighbours are far apart in absolute terms.

## DBSCAN clusters

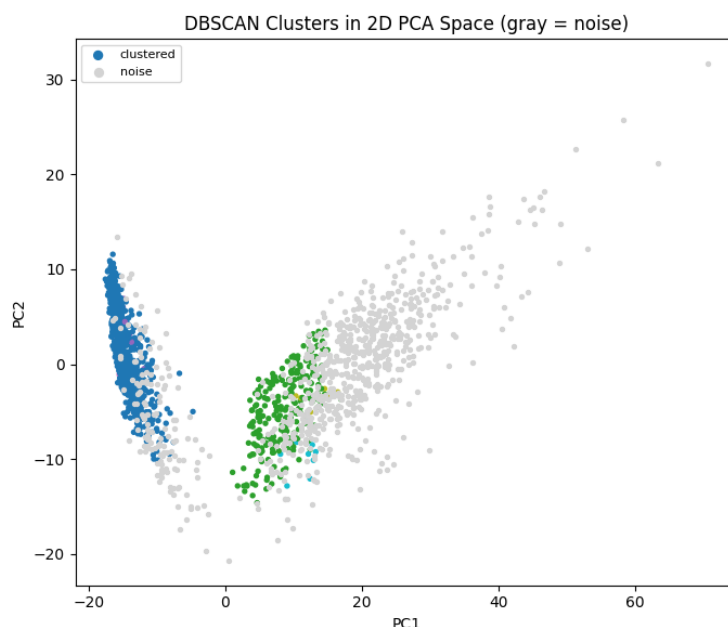


Figure 8: DBSCAN assignments in PCA space. Grey points were labelled as noise.

**Inference:** DBSCAN recovered exactly six clusters, but 39.4% of the points were discarded as noise, and the grey points are spread throughout the plot rather than confined to the edges. Dense cores are found successfully, yet the transition zones between similar activities have no density gap for the algorithm to latch onto. The dense static region is handled better than the elongated moving region, where the walking variants merge into one another.

## Dendrogram

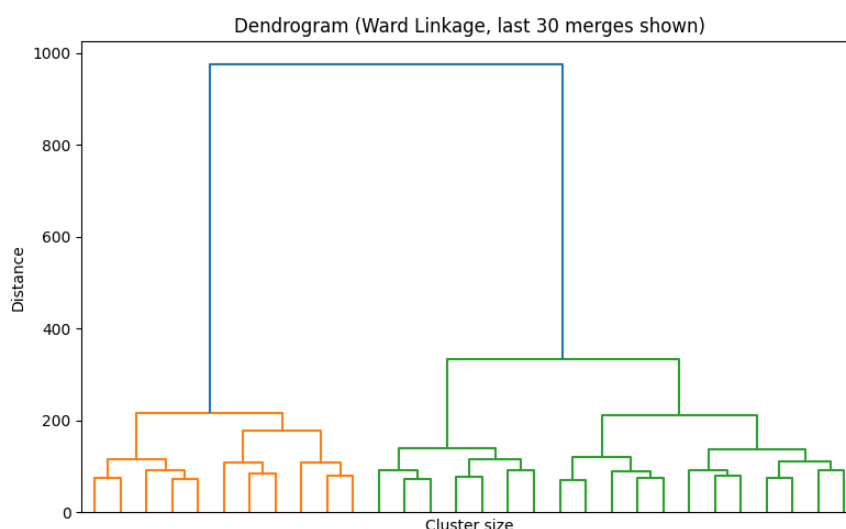


Figure 9: Ward linkage dendrogram, truncated to the last 30 merges.

**Inference:** The final merge occurs at a far greater height than any other, confirming the two dominant groups already visible in the PCA plot. Below that, merge heights increase gradually and the branches are of comparable size, which is the behaviour Ward linkage is designed to produce. Cutting the tree just below the tallest link would give two clusters, while cutting lower yields the six used here. There is no long thin chain of single point merges, so the data contains no severe outlier structure.

## Hierarchical clusters

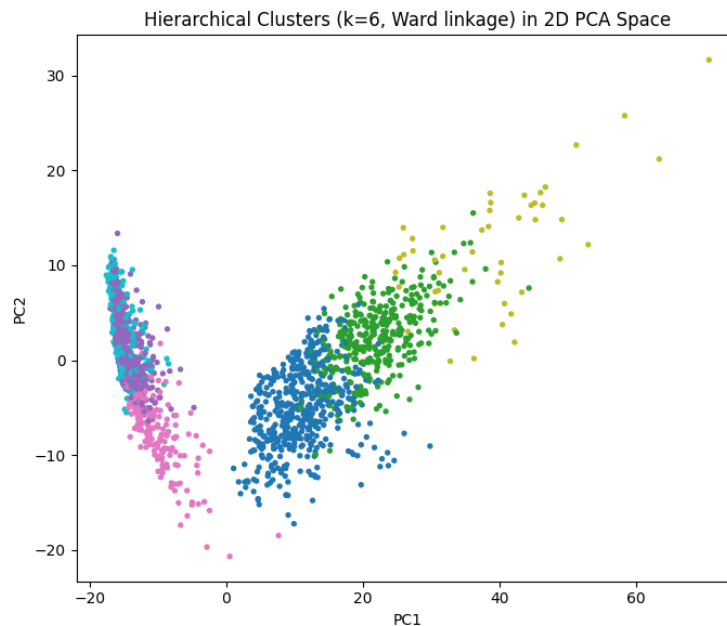


Figure 10: Agglomerative clustering with Ward linkage and  $k = 6$ , shown in PCA space.

**Inference:** The result looks broadly similar to the K-Means plot, which is unsurprising since both minimise a variance based objective. The differences appear at the borders, where Ward linkage produces slightly less regular shapes because it merges whole groups rather than reassigning individual points. This arrangement matched the true labels better than any other method tested, with ARI 0.476 and NMI 0.610.

## Comparison of evaluation metrics

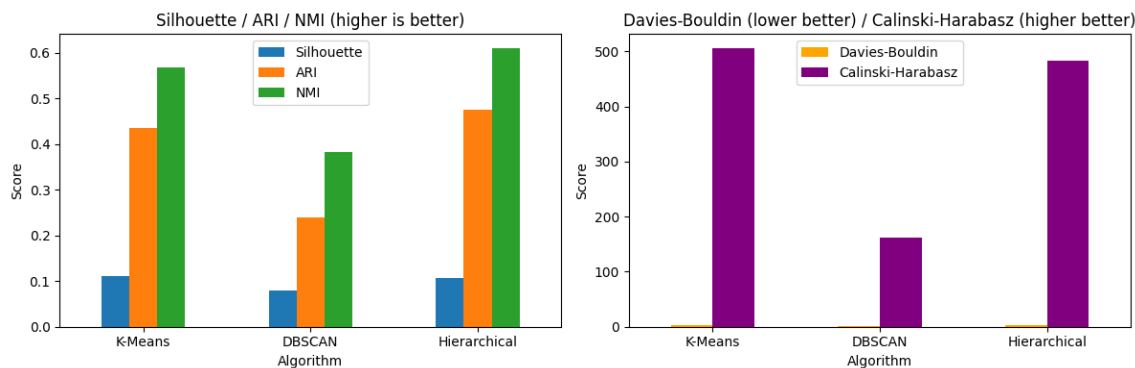


Figure 11: Internal and external validation indices for the three algorithms.

**Inference:** On the label agreement measures the ordering is Hierarchical, then K-Means, then DBSCAN, and the gap between the first two is small. DBSCAN wins on Davies–Bouldin only, which is misleading because that index is computed after its noise points are removed and therefore rewards it for refusing to classify the hardest 39.4% of the data. Calinski–Harabasz tells a much more consistent story, ranking DBSCAN far below the other two. The takeaway is that internal indices should never be read without checking how many points each algorithm actually assigned.

## 11 Discussion

**Which algorithm produced the most meaningful clusters?** Hierarchical clustering with Ward linkage agreed best with the true activities, reaching ARI 0.476 and NMI 0.610, with K-Means close behind at 0.434 and 0.568. DBSCAN trailed at 0.239 and 0.382 despite recording the lowest Davies–Bouldin score, because nearly two fifths of its points were never assigned to a cluster at all.

**Sensitivity of K-Means to  $k$ .** The dependence is strong. Silhouette fell from 0.394 at  $k = 2$  to 0.112 at  $k = 6$  and 0.088 at  $k = 8$ . Only two geometrically distinct groups exist, so any attempt to reach the six semantic classes necessarily cuts through dense regions. The value of  $k$  that looks best geometrically is therefore not the value that is correct for the problem.

**Noise handling in DBSCAN.** With  $\varepsilon = 14$  and `min_samples = 5` the algorithm found six clusters but marked 788 out of 2,000 points as noise. It did not fragment into many useless fragments, which is a positive sign, yet the boundaries between SITTING and STANDING and among the three walking variants have no density valley in 561 dimensions for it to find.

**Effect of linkage.** Only Ward linkage was used, as required by the experiment. Its variance minimising rule produces compact and similarly sized clusters and a dendrogram whose merge heights grow smoothly, quite unlike the chaining that single linkage is known to cause. Since that objective closely mirrors the one K-Means optimises, the two methods returned very similar partitions, with silhouette scores of 0.107 and 0.112 respectively.

**Which internal metric matched visual intuition best?** Calinski–Harabasz. It placed K-Means (506.0) and Hierarchical (482.5) clearly ahead of DBSCAN (161.4),

which is consistent with how the scatter plots actually look. Davies–Bouldin was the least trustworthy of the three for this dataset, for the reason already given.

**Limitations and possible improvements.** Clustering was carried out on a 2,000 row subsample for tractability, so the numbers carry some sampling variation. Applying the algorithms in a reduced PCA space, rather than on all 561 correlated features, would likely sharpen the density contrast that DBSCAN depends on. Gaussian Mixture Models would relax the spherical cluster assumption, and tuning  $\varepsilon$  and `min_samples` jointly over a grid would give DBSCAN a fairer chance than a single value read from the knee of one curve.

## 12 Conclusion

Three clustering algorithms were applied to 561 dimensional smartphone sensor data and compared using both internal and external validation indices. All three recovered the coarse division between moving and stationary activities without difficulty, but none reproduced the full six way activity structure. Ward linkage hierarchical clustering gave the closest match to the ground truth (ARI 0.476, NMI 0.610), K-Means was a close second at a fraction of the computational cost, and DBSCAN performed worst overall because high dimensional distance concentration left it unable to find density boundaries between similar activities. The experiment also showed clearly that internal indices and external indices can disagree, and that a good Davies–Bouldin score means little when 39.4 % of the data has been set aside as noise.

## 13 References

<https://github.com/sugith-2025/MachineLearning/tree/main/Experiment-8>